

```

# 1. IMPORT LIBRARIES

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from textblob import TextBlob
from wordcloud import WordCloud, STOPWORDS

# 2. LOAD DATASET
# =====
# Make sure file name is correct (NO extra space)
df = pd.read_csv("Sentiment_Analysis .csv")

# View data
print(df.head())

# 3. SENTIMENT FUNCTION

def get_sentiment(text):
    meaning = TextBlob(text)
    if meaning.sentiment.polarity > 0:
        return "Positive"
    elif meaning.sentiment.polarity < 0:
        return "Negative"
    else:
        return "Neutral"

# Apply sentiment analysis
df["Sentiment"] = df["Sentences"].apply(get_sentiment)

print(df.head())

# 4. SAVE OUTPUT

df.to_excel("Sentiment_Labeled.xlsx", index=False)

# 5. SENTIMENT COUNT

sentiment_count = df["Sentiment"].value_counts()

print(sentiment_count)

# 6. PIE CHART

plt.figure(figsize=(5,5))
plt.pie(sentiment_count, labels=sentiment_count.index, autopct="%1.2f%%")
plt.title("Sentiment Distribution (Pie Chart)")
plt.show()

# 7. BAR CHART

plt.figure()
sentiment_count.plot(kind='bar')
plt.title("Sentiment Distribution (Bar Chart)")
plt.xlabel("Sentiment")
plt.ylabel("Count")
plt.show()

# 8. BAR CHART

plt.figure()
sns.countplot(x='Sentiment', data=df)
plt.title("Sentiment Distribution (Seaborn)")
plt.show()

# 9. FILTER NEGATIVE REVIEWS

negative_reviews = df[df["Sentiment"] == "Negative"]

print(negative_reviews.head())

# Combine all negative text
negative_text = " " .join(negative_reviews["Sentences"])

```

```
negative_text = ' '.join(negative_reviews['sentences'])
```

```
# 10. WORD CLOUD
```

```
stopwords = set(STOPWORDS)  
stopwords.update(["product", "service"]) # optional
```

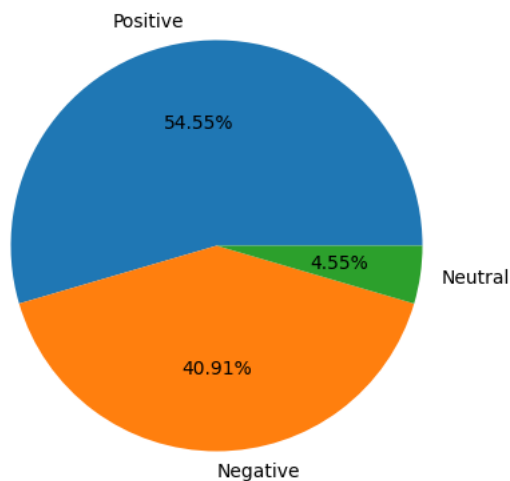
```
wc = WordCloud(  
    width=800,  
    height=400,  
    background_color='black',  
    stopwords=stopwords  
)  
.generate(negative_text)
```

```
plt.figure()  
plt.imshow(wc)  
plt.axis("off")  
plt.title("Most Common Words in Negative Reviews")  
plt.show()
```

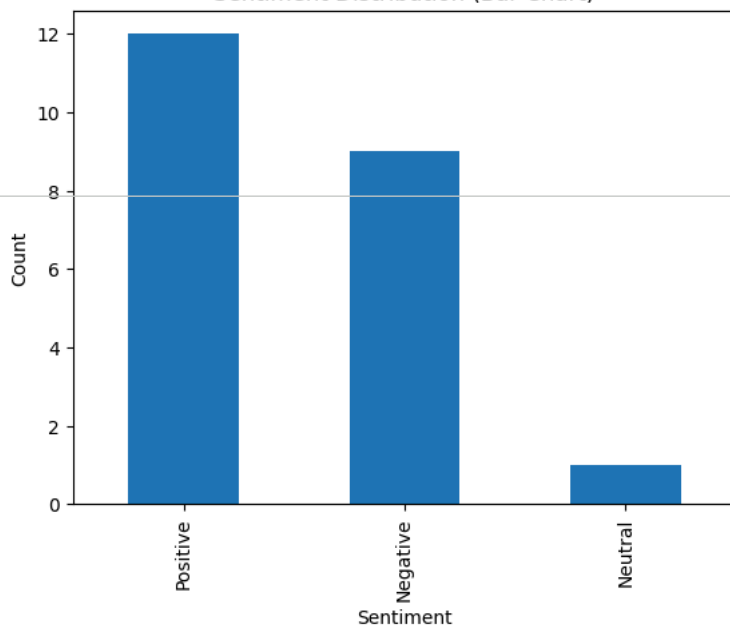
```
Sentences
0 The product's exceptional quality and performa...
1 Poor customer support and the product's subpar...
2 The service met standard requirements without ...
3 Outstanding customer service accompanied the p...
4 The product's failure to meet basic requiremen...

Sentences Sentiment
0 The product's exceptional quality and performa... Positive
1 Poor customer support and the product's subpar... Negative
2 The service met standard requirements without ... Positive
3 Outstanding customer service accompanied the p... Positive
4 The product's failure to meet basic requiremen... Negative
Sentiment
Positive    12
Negative     9
Neutral      1
Name: count, dtype: int64
```

Sentiment Distribution (Pie Chart)



Sentiment Distribution (Bar Chart)



Sentiment Distribution (Seaborn)

